

Week 1 Assignment Report
AI Foundations & Developer Environment
Name: Sheik Mohammad Riyaaz
Roll number:24JR1A05Q9

Q1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence. AI can learn from data, recognize patterns, make decisions, and solve problems. Examples include chatbots, virtual assistants, and recommendation systems.

Q2. Difference between AI, Machine Learning, Generative AI, and Agentic AI

- **Artificial Intelligence (AI):** Technology that enables machines to mimic human intelligence.
 - **Machine Learning (ML):** A subset of AI that learns from data without being explicitly programmed.
 - **Generative AI:** AI that creates new content such as text, images, music, or code.
 - **Agentic AI:** AI systems that can plan, make decisions, and perform tasks autonomously to achieve goals.
-

Q3. Explain any three open-source AI models available on Hugging Face

Llama

Llama is an open-source language model developed by Meta. It can be used for text generation, summarization, and question answering.

Mistral

Mistral is a powerful open-weight language model known for high performance and efficiency.

DeepSeek

DeepSeek is an open-source AI model used for reasoning, coding assistance, and natural language processing tasks.

Q4. Difference between Traditional Programming and AI-Assisted Development

Traditional programming follows fixed rules written by developers. The output depends entirely on the programmed instructions.

AI-assisted development uses AI tools to generate code, suggest solutions, automate tasks, and improve productivity. It helps developers work faster and more efficiently.

Q5. Explain AI Hallucinations with One Example

AI hallucination occurs when an AI system generates incorrect or fabricated information while presenting it confidently as a fact.

Example: If an AI states that a person won an award they never received, that information is a hallucination because it is not based on real facts.

```
PS C:\Users\mdriy\OneDrive\Documents\week 1 assignment> python api_task.py
Random Cat Fact:
Most cats had short hair until about 100 years ago, when it became fashionable to own cats and experiment with breeding.
PS C:\Users\mdriy\OneDrive\Documents\week 1 assignment>
```